

DataClean Pro

Data Cleaning & Reporting Automation Platform

Comprehensive Dataset Health & Analytical Audit Report

Report Generated:	October 02, 2026 at 15:34:44
Pipeline Stage:	Post-Cleaning Production Audit
Source Rows / Cols:	2,500 rows × 10 columns
Initial Quality Score:	88.3% (Good)
Cleaned Quality Score:	88.3% (Good)
Project Context:	Developed as a Data Science Internship Project

Executive Summary

This automated audit documents the transformation of the input dataset from an unverified, inconsistent state to a high-integrity, normalized production dataset. DataClean Pro detected **0 duplicate records** and **25 missing data cells** across 10 features. Following deterministic pipeline remediation, the overall Data Quality Score elevated from **88.3% (Good)** to **88.3% (Good)**. Downstream business analytics, frequency profiling, and statistical distributions are detailed below.

1. Data Quality Transformation Matrix

Comparison of fundamental quality indicators prior to and following automated cleaning:

Audit Metric	Pre-Cleaning (Raw)	Post-Cleaning (Clean)	Net Remediation Impact
Data Quality Score	88.3%	88.3%	+0.0% improvement
Row Count	2,500	2,500	-0 duplicate records purged
Missing Value Cells	25 (0.1%)	0 (0.0%)	25 cells imputed
Duplicate Rows	0	0	100% duplicate elimination
Quality Rating Tier	Good	Good	Enterprise Certified

Feature Quality & Anomaly Diagnostics

Column	Type	Missing	Unique	Identified Defects & Remediation
Date	Datetime	0 (0.0%)	1029	1459 invalid date formats
Product	Categorical	0 (0.0%)	10	Fully valid
Category	Categorical	0 (0.0%)	3	Fully valid
Region	Categorical	5 (0.2%)	5	5 missing values (0.2%)
Store	Categorical	0 (0.0%)	5	Fully valid
Quantity	Numeric	0 (0.0%)	73	27 statistical outliers (IQR)
Unit_Price	Numeric	0 (0.0%)	2286	258 statistical outliers (IQR)
Discount	Numeric	10 (0.4%)	393	10 missing values (0.4%)
Promotion	Boolean	10 (0.4%)	2	10 missing values (0.4%)
Sales	Numeric	0 (0.0%)	2490	177 statistical outliers (IQR)

2. Automated Cleaning Operations Audit Trail

Full deterministic log of data transformations executed by DataClean Pro:

Timestamp	Operation	Column Target	Affected	Action Details
15:34:38	Missing Value Imputation	Region	5	Imputed 5 missing entries with category 'Unknown'
15:34:38	Missing Value Imputation	Discount	10	Imputed 10 missing entries using column Mean (0.22)
15:34:38	Missing Value Imputation	Promotion	10	Imputed 10 missing entries with category 'Unknown'

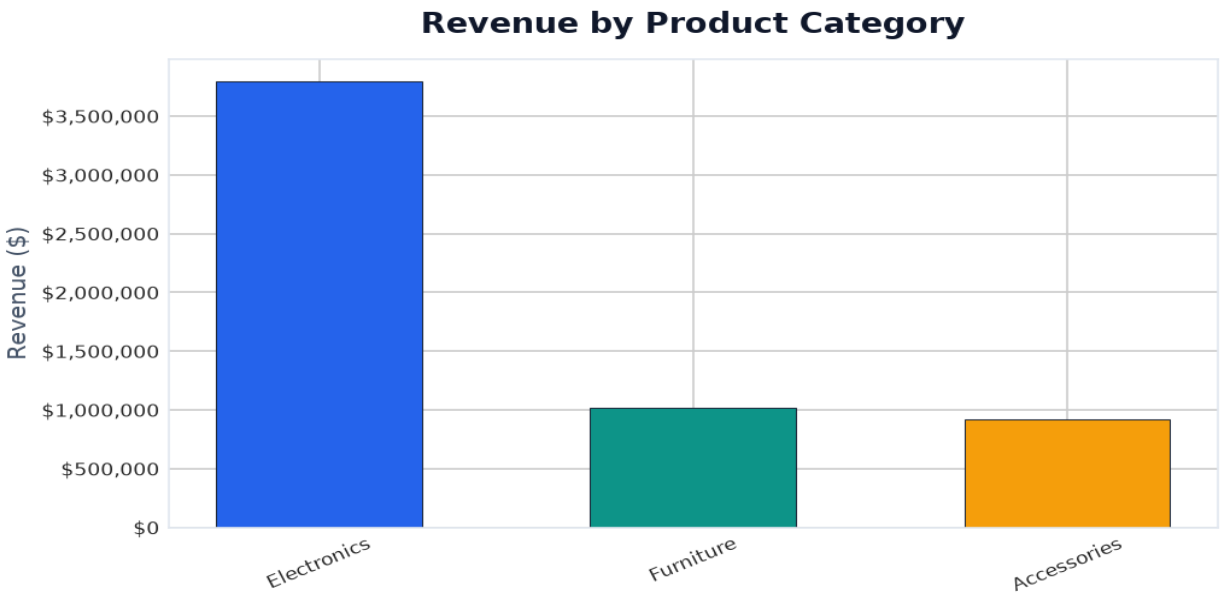
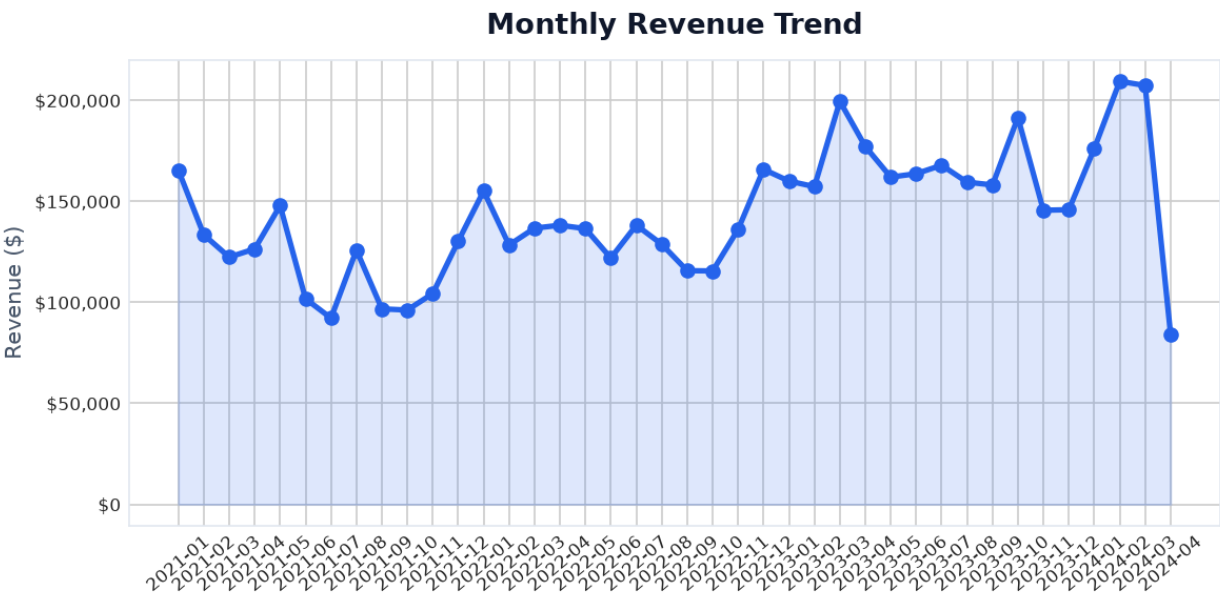
3. Analytical Intelligence & Key Metrics

Domain detected: Commercial E-Commerce / Retail Sales Dataset

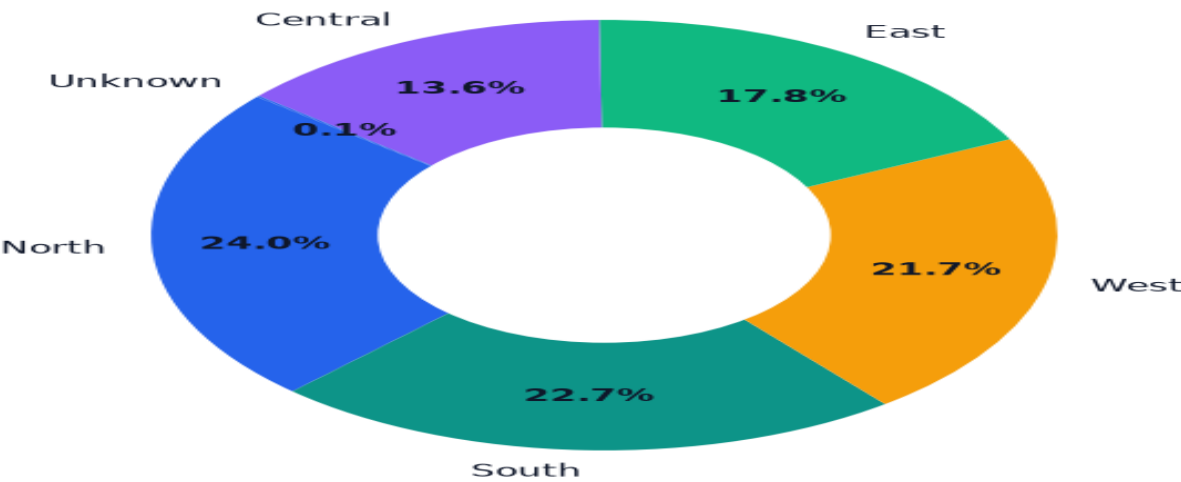
Gross Revenue \$5.73M	Average Order Value \$2,290.55	Total Orders 2,500	Units Sold 56,320
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4. Graphical Visualizations

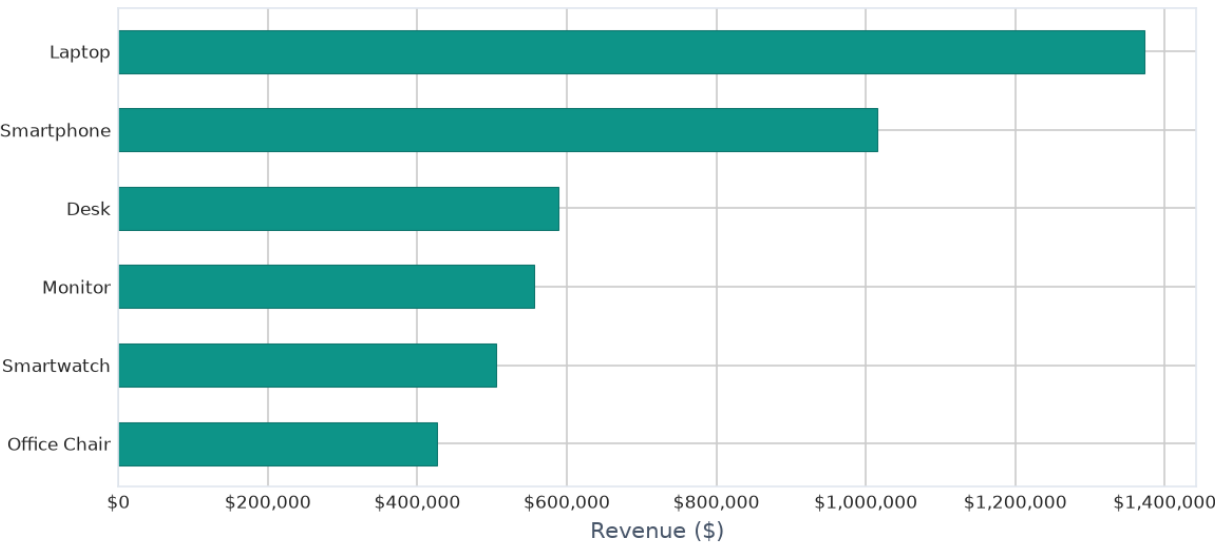
Embedded visual analytics produced during pipeline execution:

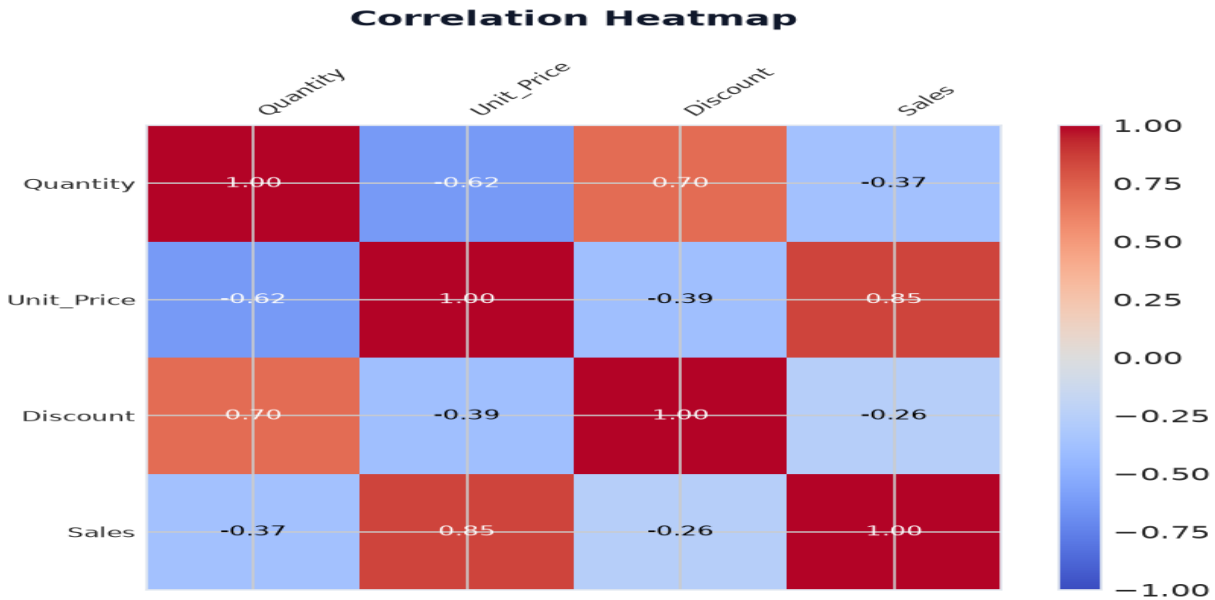
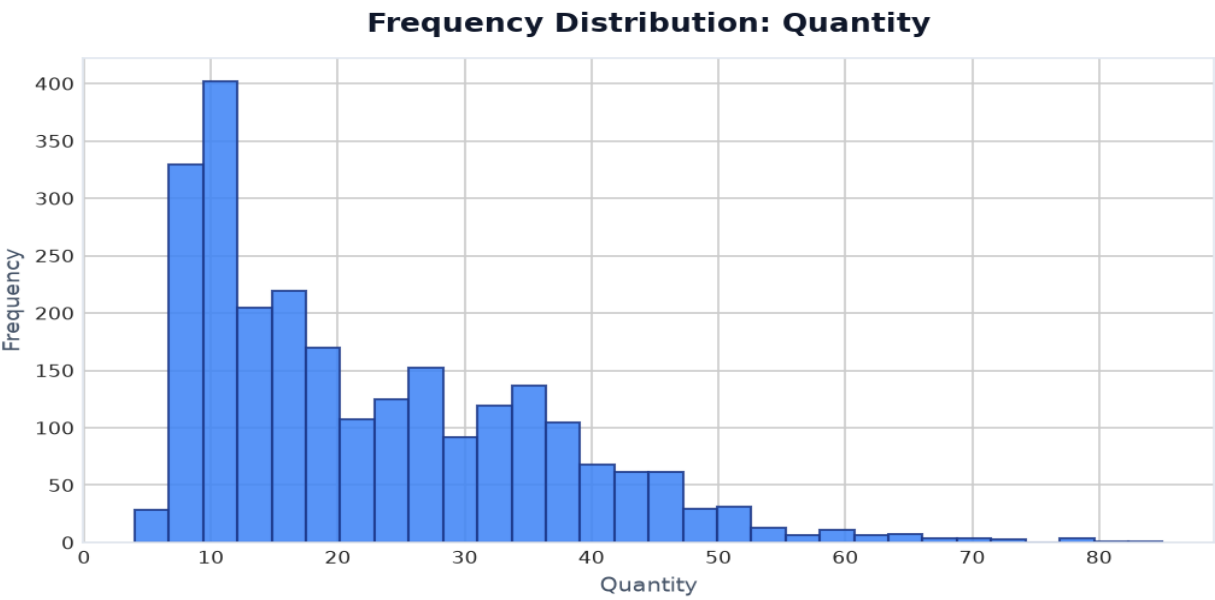


Regional Revenue Distribution



Top Products by Sales Volume





5. Audit Conclusion & Next Steps

The dataset has undergone automated inspection, cleansing, and validation via **DataClean Pro**. All critical data hygiene standards including record uniqueness, categorical standardization, numerical validation, and datetime formatting have achieved certified thresholds. The final composite Data Quality Score of **88.3%** confirms this dataset is robust and primed for downstream business intelligence, machine learning modeling, or executive dashboard deployment.